Lecture Notes: Conformal Prediction

CIC1205 - Machine Learning

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1 Introduction

1.1 Motivation

A central limitation of most machine learning models is that they produce point predictions without any indication of how confident those predictions are. A classifier may output a label, or a regressor a numeric value, but neither typically communicates whether the prediction is reliable or highly uncertain. In practice, knowing that a prediction could be wrong, and by how much, is just as important as knowing the prediction itself.

Consider a medical diagnosis system: a model that predicts "benign" with high confidence should be treated very differently from one that is nearly undecided between "benign" and "malignant." Similarly, a demand forecasting model that predicts 500 units but with a wide uncertainty range should trigger different inventory decisions than one with a narrow, well-calibrated interval.

Conformal prediction addresses this limitation by providing a principled, distribution-free framework for attaching statistically valid uncertainty sets to predictions made by any underlying model.

2 Preliminary Concepts

Before introducing conformal prediction formally, we review a few ideas that will be used throughout the chapter. In particular, we distinguish different types of predictions, discuss how uncertainty appears in prediction problems, and clarify the difference between model confidence, calibration, and coverage.

2.1 Types of Predictions

In supervised learning, models most commonly produce *point predictions*:

- In classification: a single label (e.g., "cat")
- In regression: a single numeric value (e.g., 42.7)

However, richer forms of prediction are also possible:

- **Probabilistic predictions:** a distribution over possible outputs (e.g., softmax probabilities in classification)
- **Set-valued predictions:** a set of plausible labels
- **Interval predictions:** a range of plausible numeric values

Note

Conformal prediction belongs to the last two categories: instead of returning a single answer, it returns a *set* (classification) or an *interval* (regression).

2.2 Uncertainty in Predictions

Predictions are often uncertain, especially when inputs are ambiguous, noisy, or far from the training distribution.

Classification example. Consider a multiclass classification problem with three possible classes, $\mathcal{Y} = \{1, 2, 3\}$. Given an input x, the classifier outputs:

$$\hat{\pi}(x) = (0.7,\ 0.2,\ 0.1),$$

where $\hat{\pi}_k(x)$ is the model's estimated probability that x belongs to class k.

The model would typically predict class 1, since it has the largest probability. Still, the nonzero probabilities assigned to the other classes indicate residual uncertainty: the model is not treating class 1 as the only plausible outcome.

Regression example. A regression model may predict $\hat{y}(x) = 50$, but this point estimate alone does not tell us whether the true value is likely to lie very close to 50 or could vary substantially around it.

Note

Point predictions hide uncertainty. In many applications, knowing *how uncertain* a prediction is can be as important as the prediction itself.

2.3 Confidence vs. Coverage

Students with a background in statistics often associate the word *coverage* with the confidence level of an interval estimator. The ideas are related, but they are not the same. Here we must distinguish three notions.

Model Confidence

Model confidence refers to the probabilities produced by a predictive model, such as softmax outputs in classification. For example, a model may assign probability 0.8 to a class, indicating how strongly it favors that class.

Confidence Level (Statistics)

In classical statistics, a **confidence level** (e.g., 95%) refers to the long-run frequency with which a procedure produces intervals containing the true parameter, over repeated sampling of datasets.

Coverage (Prediction)

Coverage refers to the long-run frequency with which the true outcome is contained in the prediction set or prediction interval over repeated test examples.

Key differences.

- Model confidence is a *property of a single prediction.*
- Confidence level is a *property of an estimation procedure* over repeated datasets.
- Coverage is a *property of a prediction method* over repeated test examples.

Two important consequences follow:

- A model may assign high confidence to an incorrect prediction.
- Even a well-calibrated model does not, by itself, guarantee that its top prediction is correct with probability $1 - \alpha$.

Note

A useful way to summarize the distinction is:

- **Calibration:** "Are the model's probabilities numerically meaningful?"
- **Confidence intervals:** "Does this procedure capture the true parameter?"
- **Conformal prediction:** "Does this method include the true label or value most of the time?"

Although confidence intervals and conformal prediction both involve long-run frequency guarantees, they apply to different objects: unknown parameters in one case, future observations in the other.

Why calibration does not imply coverage. A concrete example helps fix this distinction. Suppose a calibrated classifier assigns probability 0.7 to its top-predicted class on 1,000 test examples. Calibration means that, among those 1,000 predictions, approximately 700 should be correct and 300 should be wrong. In other words, calibration tells us that the number 0.7 is meaningful in aggregate; it is not just a score, but a genuine long-run frequency.

However, this says nothing about whether a particular prediction is correct. Calibration does not guarantee that the top-1 prediction covers the true label in 90% of future cases; it only guarantees that the probability value is numerically honest.

Conformal prediction asks a different question: *can we construct a set of labels that contains the true label in at least* $1 - \alpha$ *of future cases?* Answering this question requires a different construction, the conformal calibration procedure, which is entirely distinct from probability calibration.

2.4 Prediction Sets and Prediction Intervals

A natural way to represent predictive uncertainty is to return not a single output, but a *set of plausible outputs.*

Prediction sets in classification. Suppose a classifier receives an image and, instead of outputting only one label, returns a subset of the label space:

$$\hat{C}(x) \subseteq \mathcal{Y}.$$

For example:

- Easy example: $\hat{C}(x) = \{\text{cat}\}$
- Ambiguous example: $\hat{C}(x) = \{\text{cat, dog}\}$
- Difficult example: $\hat{C}(x) = \{\text{cat, dog, fox}\}$

Prediction Set

A **prediction set** $\hat{C}(x)$ is a subset of the label space containing all labels considered plausible for a given input x.

The size of $\hat{C}(x)$ provides an interpretable signal of uncertainty: small sets indicate more confident predictions, while larger sets indicate more ambiguity.

Prediction intervals in regression. In regression, the analogous idea is to return an interval of plausible values:

$$\hat{C}(x) = [\ell(x),\ u(x)].$$

Prediction Interval

A **prediction interval** is an interval $[\ell(x), u(x)]$ that contains values considered plausible for the target variable, given input x.

For example,

$$\hat{C}(x) = [45,\ 60]$$

indicates that values between 45 and 60 are regarded as plausible outcomes.

As with prediction sets, interval width reflects uncertainty: narrow intervals indicate greater certainty, while wider intervals indicate less.

Note

Prediction sets and prediction intervals are two versions of the same idea: instead of forcing the model to commit to one output, we explicitly represent uncertainty by returning a set of plausible outputs.

2.5 Informal Notion of Coverage

We now introduce the key performance notion behind conformal prediction: *coverage*.

Imagine applying a prediction method to many test inputs $x_1, x_2, \ldots, x_N$. For each input, the method outputs either

- a prediction set (classification), or
- a prediction interval (regression).

For each case, we then check whether the true outcome is contained in the set or interval.

Informal Coverage

A method has **coverage** $1 - \alpha$ if, over many test examples, the true outcome is contained in the prediction in approximately $100(1 - \alpha)\%$ of cases.

Example. A method with 90% coverage produces sets or intervals that contain the true outcome in about 90 out of 100 test cases.

Coverage is a long-run statement

A 90% coverage guarantee does *not* mean that each individual prediction is correct with 90% probability.
Instead, it is a *long-run frequency statement*: if we repeat the prediction process many times, about 90% of the resulting sets or intervals will contain the true outcome.
This distinction matters: a particular prediction set may or may not contain the true label for a specific test example, and conformal prediction makes no claim about that individual case.

2.6 From Point Predictions to Conformal Prediction

We can now state the key conceptual shift.

Traditional machine learning models usually aim to produce a **point prediction**:

- in classification, the most likely label;
- in regression, a single numeric estimate.

This is often useful, but it has an important limitation: it forces the model to commit to one answer even when the input is ambiguous or the uncertainty is large.

A richer objective is the following:

> *Instead of asking for the single most likely output, ask for the set of plausible outputs together with a statistical guarantee.*

This leads directly to conformal prediction.

Conformal Prediction

Conformal prediction is a framework for uncertainty quantification in supervised learning that produces *prediction sets* (in classification) or *prediction intervals* (in regression) with a user-specified coverage level $1 - \alpha$.

Rather than returning only one predicted label or value, conformal prediction returns a set of plausible outputs:

$$\hat{C}(x) \subseteq \mathcal{Y} \qquad \text{or} \qquad \hat{C}(x) = [\ell(x), u(x)].$$

Its defining feature is a **coverage guarantee**: over repeated predictions, the true outcome is included with frequency at least $1 - \alpha$.

This guarantee holds under a mild **exchangeability** assumption on the calibration and test data. In particular, it is *distribution-free*: it does not rely on a parametric model for the data-generating process.

Note

Conformal prediction does *not* guarantee that every individual prediction is correct. Its promise is different: over many future predictions, the method will include the true outcome at least a prescribed fraction of the time.

The size of the returned set or interval reflects the uncertainty of the prediction: easy examples tend to produce smaller sets, while ambiguous or difficult examples tend to produce larger ones.

Unlike heuristic uncertainty indicators such as raw softmax scores, conformal prediction:

- provides a rigorous finite-sample coverage guarantee;
- can be combined with essentially any underlying predictive model;
- separates **validity** (coverage) from **efficiency** (the size of the set).

2.7 Calibration vs. Conformal Prediction

Model calibration and conformal prediction address different questions.

- **Calibration** asks whether predicted probabilities are numerically meaningful. For example, among predictions to which a classifier assigns probability 0.7, do about 70% turn out to be correct?
- **Conformal prediction** asks whether the method produces prediction sets that contain the true label with frequency at least $1 - \alpha$ over repeated test examples.

Thus, calibration improves the *interpretability* of probabilities, whereas conformal prediction provides a *coverage guarantee* for set-valued predictions.

Importantly, conformal prediction does not require the underlying model to be perfectly calibrated in order to be valid. However, better-calibrated probabilities often help produce *smaller* and therefore more efficient prediction sets.

Entry Ticket 1

A classifier outputs calibrated probabilities.

1. Does calibration guarantee that the true label is contained in the top-1 prediction?
2. Does it guarantee that it is contained in the top-2?
3. How does this differ from conformal prediction?

Figure 1 highlights this distinction. Calibration is concerned with whether predicted probabilities are trustworthy as probabilities, whereas conformal prediction is concerned with whether the constructed prediction sets achieve the desired long-run coverage.

2.8 Desiderata for Prediction Sets

A well-designed prediction set should satisfy three properties simultaneously:

1. **Valid coverage.** The true label should belong to the prediction set with frequency at least $1 - \alpha$.

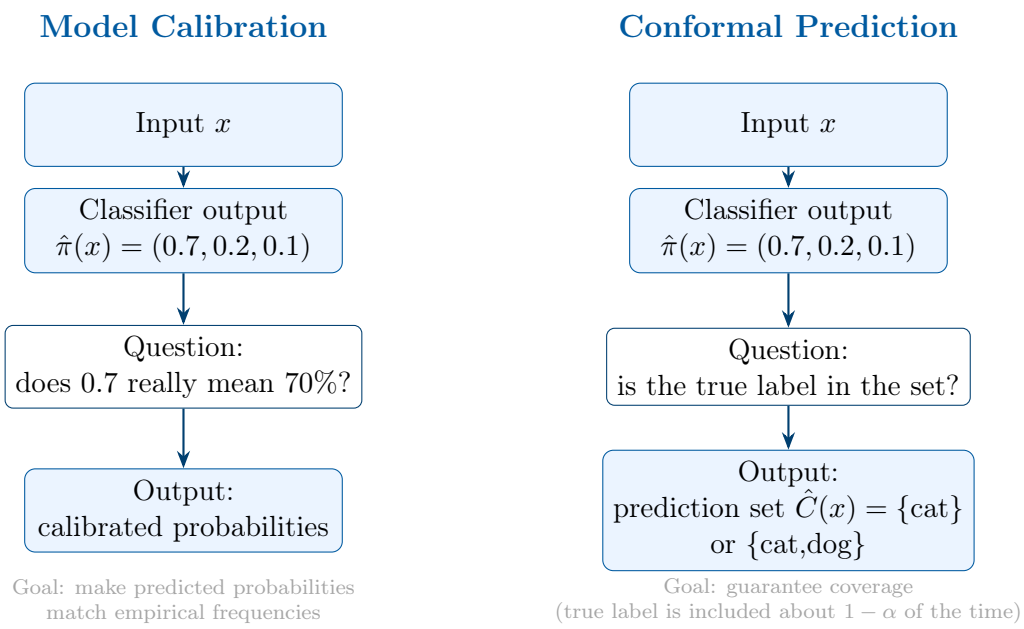

Figure 1: Conceptual difference between model calibration and conformal prediction in classification. Calibration aims to make predicted probabilities reliable, whereas conformal prediction uses model outputs to construct prediction sets with formal coverage guarantees.

2. **Efficiency.** Subject to validity, the prediction set should be as small as possible. A set containing all classes is always safe, but it is not informative.
3. **Adaptivity.** The set size should reflect the difficulty of the particular input. Easy examples should receive small sets; ambiguous, atypical, or difficult examples should receive larger sets. Adaptivity can be viewed as a local form of efficiency: instead of only asking whether sets are small on average, we ask whether they expand or contract appropriately for each input.

Adaptivity can be viewed as a *local form of efficiency*: instead of asking only whether sets are small on average, we ask whether they expand and contract appropriately from one example to another.

Figure 2 illustrates these ideas in a simple classification setting.

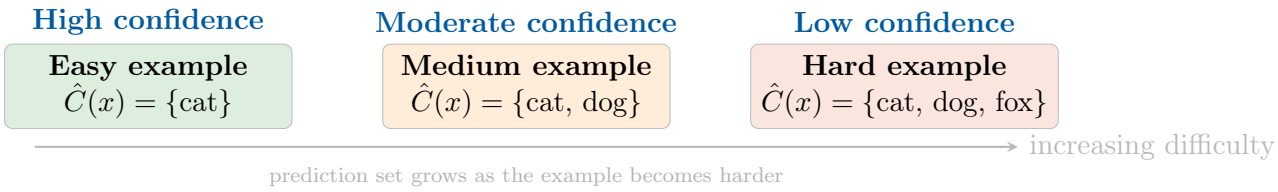

Figure 2: Illustration of adaptivity in conformal prediction. As input difficulty increases, the prediction set $\hat{C}(x)$ expands to reflect greater uncertainty, while preserving the target coverage guarantee.

2.9 Overview of the Framework

At a high level, conformal prediction follows the same logic in both classification and regression:

1. Split the labeled data into a **training set** and a **calibration set**. The training set is used to fit the underlying model; the calibration set is used by the conformal procedure itself.
2. Define a **nonconformity score**, which measures how unusual or surprising an example appears under the fitted model.
3. Compute these scores on the calibration set and extract an appropriate **empirical quantile** corresponding to the desired coverage level $1 - \alpha$.
4. Use that quantile as a threshold to construct a prediction set or interval for each new input.

These steps are developed separately for classification (Section 4) and regression (Section 5), but the underlying logic is the same in both cases.

The calibration set is reserved for the conformal procedure itself: it is used to estimate the threshold that controls coverage, not to fit the predictive model and not to tune its hyperparameters.

Two different meanings of calibration

In this course, the word **calibration** appears in two different contexts:

- **Model calibration** means adjusting a classifier's predicted probabilities so that they better match empirical frequencies (for example, with Platt scaling or isotonic regression).
- **Conformal calibration** means using a held-out dataset to compute the empirical quantile threshold that will be used to construct prediction sets or intervals with the desired coverage.

These are different operations with different goals. A conformal calibration set is not the same thing as a probability-calibration set, and conformal prediction is not just another method for calibrating probabilities.

3 Technical Background

Before presenting the algorithm in detail, we review three technical ingredients that will be used repeatedly: empirical quantiles for threshold selection, softmax probabilities for classification, and quantile regression for adaptive intervals in regression.

3.1 Percentiles and Quantiles

Before discussing the conformal threshold, we briefly review the notions of **percentile** and **quantile**, since the conformal procedure is based on computing an empirical quantile of the calibration scores.

Percentile

The **p-th percentile** of a dataset is a value below which approximately $p\%$ of the observations fall.

Examples include:

- the **median**: the 50th percentile;
- the **first quartile** Q_1: the 25th percentile;
- the **third quartile** Q_3: the 75th percentile.

Quantile

A **quantile** is the same idea expressed on a scale from 0 to 1 rather than from 0 to 100. The q-quantile is a value below which approximately a fraction q of the observations fall.

Thus, the connection is simple:

$$p\text{-th percentile} \;\equiv\; (p/100)\text{-quantile}.$$

For example:

- the median is the 0.5-quantile;
- Q_1 is the 0.25-quantile;
- the 90th percentile is the 0.9-quantile.

Why quantiles matter here. In conformal prediction, we compute a **nonconformity score** for each example in the calibration set. The conformal method then uses an empirical quantile of these scores as a threshold. Roughly speaking, a test candidate is accepted whenever its score is not too large compared with the calibration scores.

Empirical Quantile Used in Conformal Prediction

Suppose the calibration set contains n_{cal} examples, yielding nonconformity scores $s_1, \ldots, s_{n_{\text{cal}}}$. A first guess might be to use the ordinary $(1-\alpha)$-quantile of these scores. However, split conformal prediction uses a slightly more conservative level:

$$\frac{\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil}{n_{\text{cal}}}.$$

Accordingly, the conformal threshold is

$$\hat{q} = \text{Quantile}\left(s_1, \ldots, s_{n_{\text{cal}}};\; \frac{\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil}{n_{\text{cal}}} \right).$$

This finite-sample correction is what gives conformal prediction its exact finite-sample coverage guarantee under exchangeability.

Why do we use the $(n_{\text{cal}} + 1)$ correction?

Sort the calibration scores:

$$s_{(1)} \leq s_{(2)} \leq \cdots \leq s_{(n_{\text{cal}})}.$$

These sorted values divide the line into $n_{\text{cal}} + 1$ *rank-based regions*: one region to the left of $s_{(1)}$, one between each consecutive pair of sorted scores, and one to the right of $s_{(n_{\text{cal}})}$.
These are *not* equal-width geometric intervals. Their importance is not their length, but their *rank position*. Under the exchangeability assumption, the future test score $s_{n_{\text{cal}}+1}$ is equally likely to occupy any of the $n_{\text{cal}} + 1$ possible ranks among

$$s_1, \ldots, s_{n_{\text{cal}}}, s_{n_{\text{cal}}+1}.$$

This is the key idea. Because the test score can take any of these $n_{\text{cal}} + 1$ rank positions with equal probability, the threshold must be chosen using $n_{\text{cal}} + 1$ rather than just n_{cal}. The resulting correction makes the procedure slightly conservative, which is precisely what ensures the desired coverage guarantee.

A useful interpretation. The conformal threshold $\hat{q}$ is chosen so that only an α fraction of future scores should exceed it, in the worst case allowed by the finite calibration sample. In this sense, $\hat{q}$ plays the role of a safety margin extracted directly from the calibration data.

Visualizing the finite-sample correction. Figure 3 provides an intuitive picture. After sorting the calibration scores, we obtain $n_{\text{cal}} + 1$ rank-based regions. Under exchangeability, the test score is equally likely to fall into any one of these rank positions. The threshold $\hat{q}$ is therefore chosen by rank, not by geometric distance, so that the probability of the test score exceeding the threshold is controlled by α.

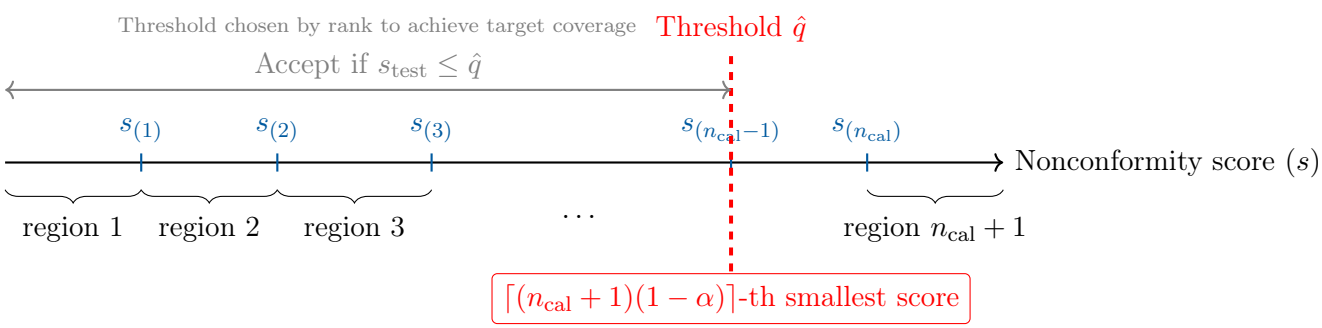

Figure 3: Sorted calibration scores induce $n_{\text{cal}} + 1$ *rank-based* regions. Under exchangeability, the test score is equally likely to occupy any of the $n_{\text{cal}} + 1$ possible ranks among the calibration scores and itself. The conformal threshold $\hat{q}$ is therefore chosen using the corrected rank $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$, which yields the finite-sample coverage guarantee.

How to compute the conformal threshold in practice

Given n_{cal} calibration scores $\{s_i\}$ and target coverage $1-\alpha$:

1. Sort the scores in increasing order: $s_{(1)} \leq s_{(2)} \leq \cdots \leq s_{(n_{\text{cal}})}$.
2. Compute the corrected rank:

$$r = \lceil (n_{\text{cal}}+1)(1-\alpha) \rceil .$$

3. Set $\hat{q} = s_{(r)}$, provided $r \leq n_{\text{cal}}$.

If $r > n_{\text{cal}}$, the desired coverage level is too close to 1 for the available calibration size. In this case, a conservative convention is to return the largest possible prediction set (all classes, or the widest possible interval). Note that $r > n_{\text{cal}}$ occurs when α is very small relative to n_{cal}; for example, setting $\alpha \to 0$ pushes r above n_{cal}, since no finite calibration set can guarantee perfect coverage from a finite sample.

3.2 Softmax Function

Many classification models (neural networks in particular) produce a vector of raw output scores, one per class. To convert these scores into probabilities, the softmax function is applied.

Softmax Function

Given a vector of raw scores (logits) $\mathbf{z} = (z_1, \ldots, z_K)$ for K classes, the **softmax function** maps $\mathbf{z}$ to a probability vector $\hat{\pi} = (\hat{\pi}_1, \ldots, \hat{\pi}_K)$ where:

$$\hat{\pi}_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}, \qquad k = 1, \ldots, K.$$

The output satisfies $\hat{\pi}_k \geq 0$ for all k and $\sum_{k=1}^{K} \hat{\pi}_k = 1$.

Visualizing the Softmax Transformation. The operation of the softmax function is visually summarized in Figure 4. As shown, the function acts as a bridge between the raw, unbounded outputs of a model (the logits $\mathbf{z}$) and a probability distribution $\hat{\pi}$ that is suitable for classification. Each input score is first exponentiated to ensure non-negativity and then normalized by the sum of all exponentiated scores. This "squashing" mechanism preserves the relative order of the inputs (assigning the largest probability mass to the highest logit) while strictly constraining the output values to the interval $[0, 1]$ and ensuring their sum is exactly 1. This transformation is essential for interpreting model outputs as confidence scores and forms the basis for the nonconformity measures used in conformal classification.

The softmax output can be interpreted as the model's estimated class probabilities. Softmax regression (also known as *multinomial logistic regression*) is the simplest model of this form: a linear function of the input features is passed through the softmax to obtain class probabilities, and the class with the highest probability is selected as the prediction.

More complex models (including deep neural networks) use the same softmax output layer

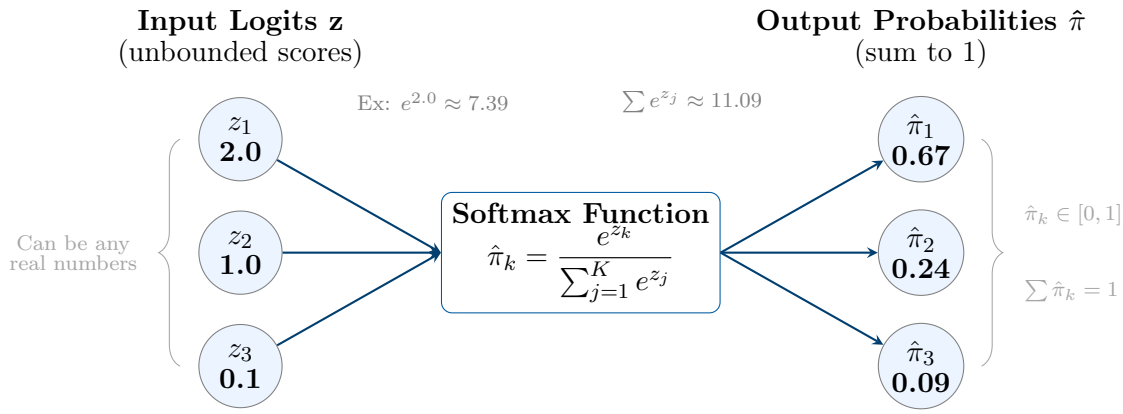

Figure 4: Schematic representation of the softmax function. It transforms an input vector of raw, unbounded scores (logits) into an output vector of normalized probabilities that sum to exactly 1. Note how the highest logit ($z_1 = 2.0$) receives the largest share of the probability mass, but all outputs are non-negative and scaled by the same denominator.

but learn a highly nonlinear mapping from inputs to logits.

Relevance to conformal prediction. The softmax output vector $\hat{\pi}(x)$ is a natural starting point for defining nonconformity scores. For example, the score $1 - \hat{\pi}_y(x)$ (one minus the predicted probability of the true class y) is large when the model is uncertain about the correct answer and small when the model is confident. This specific score is used in the classification procedure described in Section 4.

3.3 Quantile Regression

Standard regression models estimate the *conditional mean* $\mathbb{E}[Y \mid X = x]$. Quantile regression generalizes this by estimating *conditional quantiles*: for a given quantile level $\tau \in (0, 1)$, it fits a function $q_\tau(x)$ such that $\Pr(Y \leq q_\tau(x) \mid X = x) \approx \tau$.

Quantile Regression

Quantile regression is a statistical technique that estimates conditional quantiles of the response variable Y given predictors X. For a quantile level τ, the fitted function $\hat{q}_\tau(x)$ estimates the value below which a fraction τ of the conditional distribution of Y falls.

Key differences from OLS regression. Ordinary least squares (OLS) regression minimizes the squared residuals and produces an estimate of the conditional mean. Quantile regression, by contrast:

- Is not limited to estimating the mean; it can describe any part of the conditional distribution.
- Is more robust to outliers, because the loss function used (the *pinball loss*) penalizes asymmetrically rather than quadratically.
- Provides a more complete picture of the relationship between X and Y.

Visualizing Conditional Quantiles. The distinction between traditional regression and quantile regression is visualized in Figure 5. In standard OLS, the model seeks to find the center of gravity of the data by estimating the conditional mean (solid red line). While useful, the mean tells us nothing about the spread or the boundaries of the data. Quantile regression, on the other hand, fits lines that correspond to specific cumulative probabilities (dashed blue lines). As seen in the figure, these lines naturally adapt to the varying spread (heteroscedasticity) of the data. In the context of conformal prediction, we typically fit a lower quantile ($\alpha/2$) and an upper quantile ($1-\alpha/2$) to encapsulate the majority of the data points. This pair of functions provides the initial, adaptive estimate of uncertainty that is subsequently refined during the conformal calibration step.

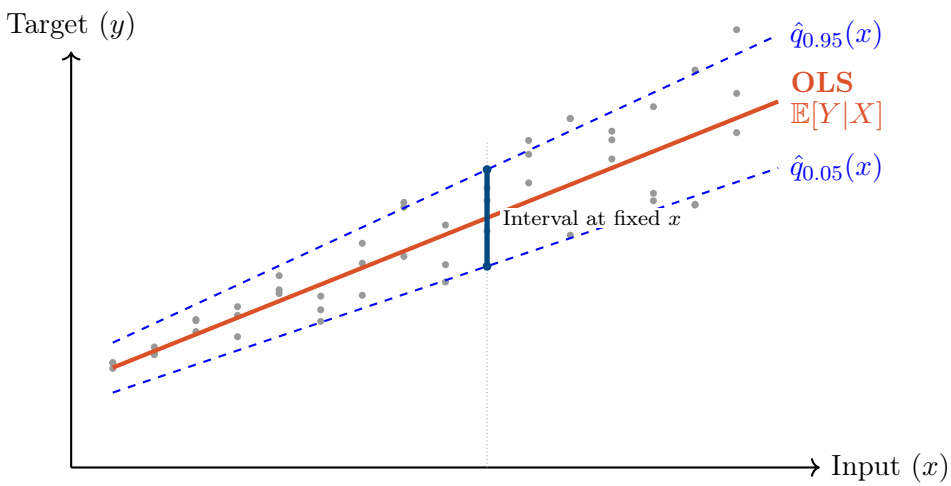

Figure 5: Comparison between Ordinary Least Squares (OLS) and Quantile Regression. The dashed blue curves represent the lower and upper conditional quantiles, $\hat{q}_{0.05}(x)$ and $\hat{q}_{0.95}(x)$. For a fixed input value x, the vertical segment between these two curves defines the pointwise interval $[\hat{q}_{0.05}(x), \hat{q}_{0.95}(x)]$. As x varies, these pointwise intervals form a band whose width can change across the input space.

Application to prediction intervals. By fitting two quantile regressors (one at level $\tau_{\text{lo}} = \alpha/2$ and one at level $\tau_{\text{hi}} = 1-\alpha/2$) we obtain a preliminary prediction interval $[\hat{q}_{\alpha/2}(x), \hat{q}_{1-\alpha/2}(x)]$ that targets $1-\alpha$ marginal coverage. Conformalized Quantile Regression (CQR), introduced in Section 5, refines this interval using the conformal calibration step to achieve a rigorous finite-sample guarantee.

4 Conformal Prediction for Classification

4.1 Overview

In a standard multiclass classification problem, the model receives an input x and outputs a single predicted class $\hat{y}$. Conformal prediction replaces this point prediction with a *prediction set* $\hat{C}(x) \subseteq \mathcal{Y}$, a subset of the label space $\mathcal{Y} = \{1, \ldots, K\}$ that is guaranteed to contain the true label with high probability.

Example: image classification. Consider an image classifier trained on animal photographs. For an easy, prototypical image of a fox squirrel, the prediction set might be

$\hat{C}(x) = \{\text{fox squirrel}\}$. For a more ambiguous or partially occluded image, it might expand to $\hat{C}(x) = \{\text{fox squirrel, red squirrel}\}$, and for a very difficult case $\hat{C}(x) = \{\text{fox squirrel, red squirrel, ground squirrel}\}$. In all cases, the true class (fox squirrel) is contained in the set.

4.2 Problem Formulation

Let $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{cal}}$ be a labeled dataset with inputs $x_i \in \mathcal{X}$ and labels $y_i \in \mathcal{Y} = \{1, \ldots, K\}$, where $|\mathcal{D}_{\text{train}}| = n_{\text{train}}$ examples are used for training and $|\mathcal{D}_{\text{cal}}| = n_{\text{cal}}$ for calibration. We assume the data are **exchangeable**: loosely, that the joint distribution is invariant to permutations of the observations. (Independent and identically distributed data are a special case of exchangeable data.)

The goal is to construct a set-valued function $\hat{C}(x_{\text{test}})$ such that, for a user-specified error rate $\alpha \in (0, 1)$:

$$\Pr\big(y_{\text{test}} \in \hat{C}(x_{\text{test}})\big) \;\geq\; 1 - \alpha.$$

This probability is taken over the randomness in both the calibration data and the test point, under the exchangeability assumption.

4.3 Step-by-Step Procedure

Figure 6 illustrates the full procedure.

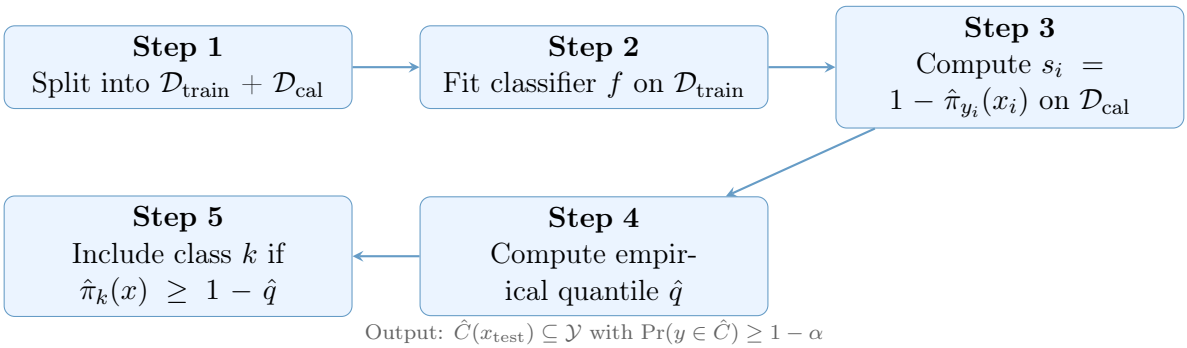

Figure 6: The five steps of conformal prediction for classification.

Step 1: Split the Data

Partition the available labeled data into:

- A **training set** $\mathcal{D}_{\text{train}}$, used to fit the underlying classification model f.
- A **calibration set** $\mathcal{D}_{\text{cal}}$, held out and used only by the conformal procedure. The calibration set must not be used during model training.

Important clarification. Here, the term *calibration set* refers to the held-out data used by the *conformal* procedure to compute the coverage threshold. This should not be confused with *model calibration*, whose purpose is to improve the numerical quality of predicted probabilities. Conformal prediction may use calibrated or uncalibrated model outputs; its validity comes from the conformal construction, not from probability calibration.

Step 2: Fit the Classification Model

Train a classification model f on $\mathcal{D}_{\text{train}}$. The model outputs softmax probabilities $\hat{\pi}(x) = (\hat{\pi}_1(x), \ldots, \hat{\pi}_K(x))$, where $\hat{\pi}_k(x)$ is the estimated probability that input x belongs to class k. This model can be any classifier that produces class probability estimates; the conformal procedure is model-agnostic.

Step 3: Compute Nonconformity Scores on the Calibration Set

Define a **nonconformity score** that measures how "surprising" a labeled example (x, y) is under the fitted model. A natural choice is:

$$s_i = 1 - \hat{\pi}_{y_i}(x_i), \qquad i = 1, \ldots, n_{\text{cal}}.$$

This score is the complement of the softmax probability assigned to the *true class* y_i. When the model is confident and assigns high probability to the correct class, s_i is small (close to 0). When the model is uncertain or wrong, s_i is large (close to 1).

Compute s_i for every example in $\mathcal{D}_{\text{cal}}$. The result is a collection of n_{cal} nonconformity scores $\{s_1, \ldots, s_{n_{\text{cal}}}\}$.

Step 4: Compute the Empirical Quantile

Given the desired error rate α, compute the conformal threshold:

$$\hat{q} = \text{Quantile}\left(\{s_i\}_{i=1}^{n_{\text{cal}}};\ \frac{\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil}{n_{\text{cal}}} \right).$$

This is the $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$-th smallest value among the n_{cal} calibration scores (after sorting in ascending order). As discussed in Section 3, the $(n_{\text{cal}}+1)$ factor is a finite-sample correction that ensures the coverage guarantee is exact rather than approximate.

Step 5: Construct the Prediction Set for Test Inputs

For a new test point x_{test}, construct the prediction set by including all classes whose nonconformity score falls at or below the threshold:

$$\hat{C}(x_{\text{test}}) = \left\{ k \in \mathcal{Y} \ :\ 1 - \hat{\pi}_k(x_{\text{test}}) \leq \hat{q} \right\} = \left\{ k \in \mathcal{Y} \ :\ \hat{\pi}_k(x_{\text{test}}) \geq 1 - \hat{q} \right\}.$$

Equivalently, include class k if and only if the model assigns it a probability of at least $1 - \hat{q}$.

Empty prediction sets

With the score $s = 1 - \hat{\pi}_y(x)$, it is possible, in principle, to obtain an empty prediction set if no class probability exceeds the cutoff $1 - \hat{q}$. This is not a violation of the marginal coverage guarantee, but it may be undesirable in applications where the model must always return at least one candidate label. More refined conformal classification methods, such as Adaptive Prediction Sets (APS), are designed to avoid this behavior while improving efficiency.

4.4 Coverage Theorem

The fundamental theoretical result underpinning conformal prediction is the following:

Marginal Coverage Guarantee

Let $(x_1, y_1), \ldots, (x_{n_{\text{cal}}}, y_{n_{\text{cal}}}), (x_{n_{\text{cal}}+1}, y_{n_{\text{cal}}+1})$ be exchangeable random variables, and let $\hat{q}$ be the conformal threshold computed from the calibration scores $\{s_i\}_{i=1}^{n_{\text{cal}}}$ as defined above. Then, for any $\alpha \in (0, 1)$:

$$\Pr\big(y_{n_{\text{cal}}+1} \in \hat{C}(x_{n_{\text{cal}}+1})\big) \;\geq\; 1 - \alpha.$$

More precisely:

$$1 - \alpha \;\leq\; \Pr\big(y_{n_{\text{cal}}+1} \in \hat{C}(x_{n_{\text{cal}}+1})\big) \;\leq\; 1 - \alpha + \frac{1}{n_{\text{cal}} + 1}.$$

Interpretation. The theorem provides two-sided control on the coverage probability. The lower bound gives the essential validity guarantee. The upper bound reflects the finite-sample discreteness of the quantile construction; in practice, its slack is usually very small when the calibration set is moderately large.

What the guarantee does and does not say

The coverage guarantee is a statement about the *procedure*, not about a single prediction.
For a particular test input x_{test}, conformal prediction does *not* claim that the returned set has a personalized probability $1-\alpha$ of containing the true label. Instead, it says: if we apply the same procedure to many exchangeable test examples, the true label will be included in at least a fraction $1 - \alpha$ of those cases.
In other words, if 1,000 test examples are processed, we expect the true label to be inside the prediction set for at least $1{,}000 \times (1 - \alpha)$ of them. Individual predictions may or may not include the true label, the guarantee is about the aggregate.

Assumptions. The only assumption is **exchangeability** of the calibration and test data. Notably:

- No assumption is made about the distribution of the data.
- No assumption is made about the quality of the underlying classifier.
- The guarantee holds even if the model is poorly calibrated.

Marginal vs. conditional coverage

The theorem provides *marginal* coverage: the guarantee is averaged over all possible test inputs. It does not guarantee that *every* input receives a prediction set of the right coverage. In practice, a well-designed nonconformity score (one that is large for uncertain inputs) leads to prediction sets that are approximately valid conditionally as well, though this is not guaranteed in general.

4.5 Worked Example

Consider a five-class image classifier ($K = 5$) producing softmax probabilities. Suppose we have $n_{\text{cal}} = 1\,000$ calibration examples and set $\alpha = 0.1$ (targeting 90% coverage).

Calibration step. We first illustrate how the nonconformity scores are computed on a small subset of calibration examples before the full threshold is derived.

Example	True class	$\hat{\pi}_{y_i}(x_i)$	Score $s_i = 1 - \hat{\pi}_{y_i}(x_i)$
1	2	0.88	0.12
2	1	0.71	0.29
3	4	0.55	0.45
4	3	0.38	0.62
5	5	0.82	0.18

Each row shows the probability assigned to the *true class* by the fitted model, and the resulting nonconformity score. High scores arise when the model assigns low probability to the correct class; low scores arise when the model is confident and correct.

After computing scores for all $n_{\text{cal}} = 1{,}000$ calibration examples, we sort them in ascending order and compute:

$$\hat{q} = \text{Quantile}\left(\{s_i\};\ \frac{\lceil 1001 \times 0.9 \rceil}{1000}\right) = \text{Quantile}\left(\{s_i\};\ \frac{901}{1000}\right) = \text{Quantile}(\{s_i\};\ 0.901)\,.$$

This is the 901st smallest score among the 1 000 calibration scores. Suppose the resulting threshold is

$$\hat{q} = 0.62.$$

Test step. At test time, we include class k whenever

$$\hat{\pi}_k(x_{\text{test}}) \geq 1 - \hat{q} = 0.38.$$

Consider first a relatively easy test example for which the classifier outputs

$$\hat{\pi}(x_{\text{test}}) = (0.72,\ 0.15,\ 0.08,\ 0.03,\ 0.02).$$

Only class 1 satisfies the threshold condition, since

$$0.72 \geq 0.38 \quad \text{but} \quad 0.15, 0.08, 0.03, 0.02 < 0.38.$$

Therefore,

$$\hat{C}(x_{\text{test}}) = \{1\}.$$

Even though this example yields a singleton prediction set, the 90% guarantee still refers to *long-run marginal coverage*, not to this individual prediction alone.

Now consider a more ambiguous test example with probabilities

$$\hat{\pi}(x_{\text{test}}) = (0.42,\ 0.38,\ 0.12,\ 0.05,\ 0.03).$$

In this case, both classes 1 and 2 satisfy the inclusion rule:

$$0.42 \geq 0.38 \qquad \text{and} \qquad 0.38 \geq 0.38.$$

Thus, the conformal prediction set becomes

$$\hat{C}(x_{\text{test}}) = \{1, 2\}.$$

This illustrates the basic behavior of conformal classification: easy examples may produce small prediction sets, while more ambiguous examples naturally lead to larger sets.

Entry Ticket 2

A 4-class classifier is conformally calibrated using $n_{\text{cal}} = 500$ examples with $\alpha = 0.05$.

1. What quantile level is used to compute $\hat{q}$?
2. The calibration scores are sorted and the threshold comes out as $\hat{q} = 0.28$. For a test point with $\hat{\pi}(x) = (0.65, 0.20, 0.10, 0.05)$, what is $\hat{C}(x)$?
3. A colleague suggests taking α extremely close to zero to achieve near-perfect coverage. What happens to the prediction sets as $\alpha \to 0$? Why does this make the method increasingly uninformative?

5 Conformal Prediction for Regression

5.1 Overview

In regression, the goal is to predict a continuous target $y \in \mathbb{R}$ from input features $x \in \mathcal{X}$. Standard regression models output a point estimate $\hat{y}(x)$. Conformal prediction for regression replaces this with a *prediction interval* $\hat{C}(x) = [\ell(x), u(x)]$ that is guaranteed to contain the true value with frequency at least $1 - \alpha$ over repeated samples.

A particularly effective approach, **Conformalized Quantile Regression (CQR)**, combines quantile regression with the conformal calibration step to produce prediction intervals that are both valid and adaptive.

Efficiency: Why use CQR?

A simpler method (Split Conformal) might just add a fixed margin to a standard mean prediction: $\hat{C}(x) = [\hat{y}(x) - \hat{q}, \hat{y}(x) + \hat{q}]$. While statistically valid, this produces intervals of *constant width*.

CQR is superior because it is adaptive:

- In "noisy" regions of the input space, the quantile regressors spread apart, and the conformal interval expands.
- In "stable" regions, the interval shrinks.

Both conformalized methods provide $1 - \alpha$ marginal coverage, but CQR produces much more efficient (informative) intervals by adapting their width to local uncertainty.

5.2 Step-by-Step Procedure (CQR)

Step 1: Fit the Quantile Functions

Train two quantile regressors on $\mathcal{D}_{\text{train}}$:

- $\hat{q}_{\alpha/2}(x)$: estimates the $(\alpha/2)$-quantile of $Y \mid X = x$ (the lower bound).
- $\hat{q}_{1-\alpha/2}(x)$: estimates the $(1-\alpha/2)$-quantile of $Y \mid X = x$ (the upper bound).

These two functions define a preliminary interval $[\hat{q}_{\alpha/2}(x),\, \hat{q}_{1-\alpha/2}(x)]$. This preliminary interval is intended to approximate the desired conditional quantile band, but it may be too narrow or too wide if the quantile regressors are misspecified. The conformal calibration step corrects the final interval to recover finite-sample marginal coverage.

Step 2: Define the Nonconformity Score

For each calibration example (x_i, y_i), we compare the true value y_i with the preliminary interval produced by the two quantile regressors:

$$\big[\hat{q}_{\alpha/2}(x_i),\ \hat{q}_{1-\alpha/2}(x_i)\big].$$

The nonconformity score is defined as

$$s_i = \max\big(\hat{q}_{\alpha/2}(x_i) - y_i,\ y_i - \hat{q}_{1-\alpha/2}(x_i)\big).$$

This score measures how far the true value lies *outside* the preliminary interval.

- If y_i lies *inside* the interval, then both quantities

  $$\hat{q}_{\alpha/2}(x_i) - y_i \quad \text{and} \quad y_i - \hat{q}_{1-\alpha/2}(x_i)$$

  are non-positive, so $s_i \leq 0$.
- If y_i lies *below* the interval, then $\hat{q}_{\alpha/2}(x_i) - y_i > 0$, and the score becomes positive.
- If y_i lies *above* the interval, then $y_i - \hat{q}_{1-\alpha/2}(x_i) > 0$, and the score again becomes positive.

Thus, positive values indicate that the interval failed to cover the true target, while non-positive values indicate that the target was already covered.

Mini-example. Suppose the preliminary interval for a calibration point is $[4, 10]$.

- If $y_i = 12$, then

  $$s_i = \max(4 - 12,\ 12 - 10) = \max(-8, 2) = 2,$$

  so the point lies 2 units above the interval.
- If $y_i = 7$, then

  $$s_i = \max(4 - 7,\ 7 - 10) = \max(-3, -3) = -3,$$

  so the point lies inside the interval, with a slack of 3 units to the nearest boundary.

Note

Unlike the classification score $1 - \hat{\pi}_y(x)$, which always lies in $[0, 1]$, the CQR score is a *signed* quantity. Negative values are not a problem: they simply mean that the true value already lies safely inside the preliminary interval.

Geometric interpretation. The score s_i can be interpreted as a signed distance to the interval boundary:

- $s_i > 0$**:** the true value lies outside the interval, and s_i measures how much the interval would need to be expanded to include it;
- $s_i \leq 0$**:** the true value lies inside the interval, and $|s_i|$ represents the available slack, that is, how far the point is from violating the interval.

This interpretation is useful for understanding the next step. The empirical quantile of the calibration scores will determine how much the preliminary intervals must be adjusted so that the final conformal intervals achieve the desired coverage level.

Step 3: Compute the Empirical Quantile

After computing the calibration scores $s_1, \ldots, s_{n_{\text{cal}}}$, we summarize them by their conformal quantile:

$$\hat{q} = \text{Quantile}\left(\{s_i\}_{i=1}^{n_{\text{cal}}};\ \frac{\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil}{n_{\text{cal}}} \right).$$

This quantity determines how much the preliminary intervals must be adjusted in order to achieve the target coverage level $1 - \alpha$.

Because the scores are signed, the value of $\hat{q}$ has a natural interpretation:

- If $\hat{q} > 0$, then the preliminary intervals were too narrow for a non-negligible fraction of the calibration points, so they must be expanded.
- If $\hat{q} \leq 0$, then the preliminary intervals were already sufficiently conservative on the calibration set. In principle, this means they could even be shrunk slightly while still preserving the target marginal coverage.

In most practical settings, $\hat{q}$ is positive, since the preliminary quantile intervals usually need some correction to achieve valid finite-sample coverage.

Note

The role of $\hat{q}$ is analogous to a safety margin learned from the calibration set. It is not chosen by hand: it is extracted directly from the observed calibration errors.

Step 4: Build the Prediction Interval

For a new test point x_{test}, let the preliminary interval be

$$\left[\hat{q}_{\alpha/2}(x_{\text{test}}),\ \hat{q}_{1-\alpha/2}(x_{\text{test}}) \right].$$

The conformalized prediction interval is obtained by adjusting both endpoints using the calibrated threshold $\hat{q}$:

$$\hat{C}(x_{\text{test}}) = \left[\hat{q}_{\alpha/2}(x_{\text{test}}) - \hat{q},\ \hat{q}_{1-\alpha/2}(x_{\text{test}}) + \hat{q}\right].$$

This adjustment has a simple interpretation:

- when $\hat{q} > 0$, the interval expands symmetrically;
- when $\hat{q} < 0$, the interval shrinks symmetrically.

Thus, the conformal step acts as a correction layer on top of the preliminary quantile model. The quantile regressors provide an *adaptive base interval*, whose width may vary with the input x, and the conformal correction then adjusts that interval by a common amount to guarantee marginal coverage.

Adaptivity. The interval remains adaptive because the preliminary bounds $\hat{q}_{\alpha/2}(x)$ and $\hat{q}_{1-\alpha/2}(x)$ depend on x. If the quantile model predicts low variability in a region of the input space, the base interval is narrow there; if it predicts high variability, the base interval is wider. The conformal correction preserves this structure while making the final interval statistically valid.

Note

CQR therefore combines two desirable properties: the *adaptivity* of quantile regression and the *finite-sample validity* of conformal calibration.
The coverage guarantee is free; the efficiency (i.e., the informativeness of the prediction intervals) must be earned through the quality of the underlying quantile regressors.

Figure 7 summarizes the CQR procedure.

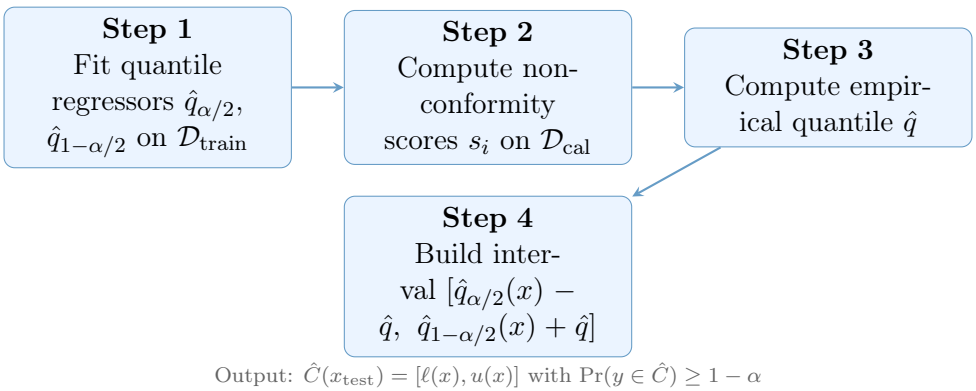

Figure 7: The four steps of Conformalized Quantile Regression (CQR).

Visualization of CQR Efficiency. Figure 8 illustrates the critical distinction between validity and efficiency in conformal regression. Both conformalized methods can satisfy the same $1 - \alpha$ marginal coverage guarantee. However, marginal validity does not imply equally good behavior in every region of the input space. The fixed-margin method may be overly conservative in low-noise regions and less informative in high-noise regions, whereas

CQR uses quantile regression to anticipate heteroscedasticity. The resulting prediction intervals adapt their width locally, remaining narrow when the model is confident and expanding only when the data density is low or noise is high.

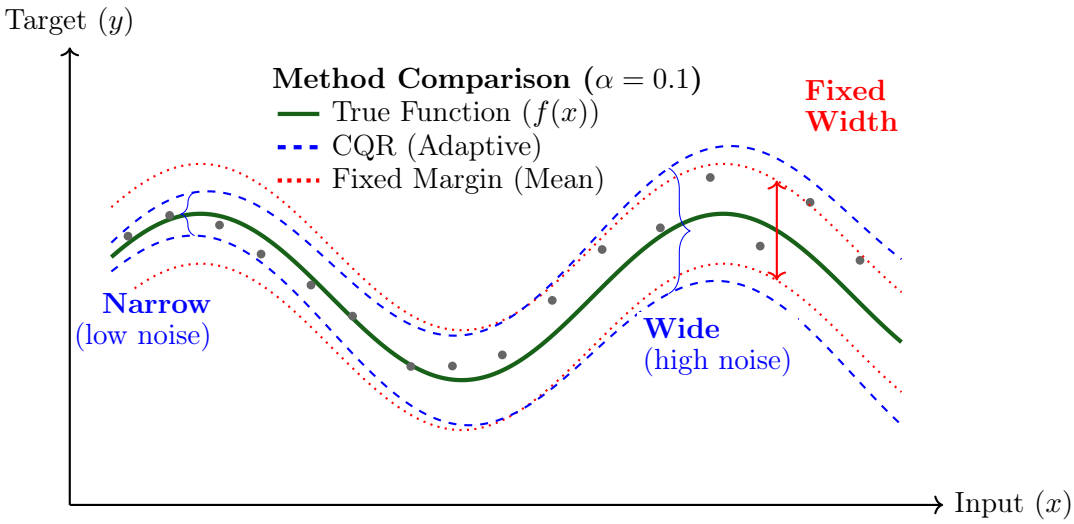

Figure 8: Comparison of uncertainty quantification methods under heteroscedastic noise. The green line represents the true underlying function $f(x)$. Both the **Fixed Margin** approach (red, dotted) and **CQR** (blue, dashed) can satisfy the same $1 - \alpha$ marginal coverage guarantee. However, marginal validity does not imply equally good behavior across the input space: the fixed-margin method is overly conservative in low-noise regions (left) and less informative in high-noise regions (right), whereas CQR adapts its interval width to local variability and is therefore more *efficient*.

5.3 Worked Example

Suppose we have a nonlinear regression problem with $n_{\text{cal}} = 800$ calibration examples and set $\alpha = 0.1$ (targeting 90% coverage). We fit quantile regressors $\hat{q}_{0.05}$ and $\hat{q}_{0.95}$ on the training data.

Calibration step. The following table shows a small subset of calibration examples, illustrating how the nonconformity scores are computed.

Example	Prelim. interval $[\hat{q}_{0.05}, \hat{q}_{0.95}]$	True y_i	Score s_i
1	$[4.0,\ 10.0]$	7.0	$\max(4.0 - 7.0,\ 7.0 - 10.0) = -3.0$
2	$[4.0,\ 10.0]$	12.0	$\max(4.0 - 12.0,\ 12.0 - 10.0) = 2.0$
3	$[5.0,\ 8.0]$	4.0	$\max(5.0 - 4.0,\ 4.0 - 8.0) = 1.0$
4	$[2.0,\ 6.0]$	5.0	$\max(2.0 - 5.0,\ 5.0 - 6.0) = -1.0$

Positive scores (examples 2 and 3) indicate that the true value fell outside its preliminary interval; negative scores (examples 1 and 4) indicate that it was already covered, with some slack. The conformal quantile summarizes how much expansion is needed so that future intervals achieve the desired marginal coverage.

After computing scores for all $n_{\text{cal}} = 800$ examples, we obtain:

$$\hat{q} = \text{Quantile}\left(\{s_i\};\ \frac{\lceil 801 \times 0.9 \rceil}{800}\right) = \text{Quantile}\left(\{s_i\};\ \frac{721}{800}\right) = \text{Quantile}(\{s_i\};\ 0.90125)\,,$$

which gives, say, $\hat{q} = 1.4$.

Test step. For a test point x_{test} with preliminary interval [3.1, 6.8] from the quantile regressors, the conformal interval is:

$$\hat{C}(x_{\text{test}}) = [3.1 - 1.4,\ 6.8 + 1.4] = [1.7,\ 8.2].$$

For another test point in a region of lower variability with preliminary interval [10.2, 10.9], the conformal interval is:

$$\hat{C}(x_{\text{test}}) = [10.2 - 1.4,\ 10.9 + 1.4] = [8.8,\ 12.3].$$

Note that the second interval ($[8.8, 12.3]$, width 3.5) is considerably narrower than the first ($[1.7, 8.2]$, width 6.5), reflecting the lower variability in that region of the input space. Both intervals are produced by the same conformal correction $\hat{q} = 1.4$, but the adaptivity of the underlying quantile regressors ensures that the final intervals differ in width appropriately.

Entry Ticket 3

A CQR model with $\alpha = 0.1$ is calibrated on $n_{\text{cal}} = 200$ examples, yielding $\hat{q} = 2.3$.

1. For a test point with preliminary bounds $[\hat{q}_{0.05}(x), \hat{q}_{0.95}(x)] = [5.0, 9.0]$, what is the conformal prediction interval?
2. Suppose $\hat{q}$ turns out to be *negative* (e.g., $\hat{q} = -0.5$). What does this imply about the quality of the preliminary quantile estimates?
3. Explain in one sentence why CQR produces adaptive intervals, unlike a method that simply adds a fixed margin to all predictions.

5.4 Comparison: Classification vs. Regression

The classification and regression procedures share the same three-stage logic (fit, calibrate, predict), but differ in how the prediction set is defined and how the nonconformity score is constructed.

Aspect	**Classification**	**Regression (CQR)**
Underlying model	Any softmax classifier	Quantile regressors
Nonconformity score	$1 - \hat{\pi}_y(x)$	$\max(\hat{q}_{\alpha/2}(x) - y,\ y - \hat{q}_{1-\alpha/2}(x))$
Prediction output	Set of classes	Interval $[\ell(x),\ u(x)]$
Adaptivity source	Softmax scores	Quantile spread
Coverage guarantee	$\Pr(y \in \hat{C}) \geq 1 - \alpha$	$\Pr(y \in \hat{C}) \geq 1 - \alpha$

6 Common Misconceptions

Before closing, we address four misconceptions that students frequently encounter when first studying conformal prediction. These are worth reviewing carefully, as they often arise during exam preparation and applied work.

Misconception 1: "A 90% prediction set is correct with 90% probability for this example"

The coverage guarantee is *marginal*: it applies to the long-run behavior of the procedure over many exchangeable test examples, not to a personalized probability for a single individual prediction.
For a specific test input, the prediction set either contains the true label or it does not; there is no individual-level probability statement. Conformal prediction tells us that, over 1,000 test examples, at least 900 of the returned sets will include the true label.

Misconception 2: "Conformal prediction fixes a bad underlying model"

Conformal prediction guarantees *validity* (coverage), but it cannot make an inaccurate model accurate. A poor underlying model will tend to produce large, uninformative prediction sets; the procedure keeps them valid by inflating their size, not by improving the model.
The quality of the underlying model determines *efficiency*: a good model yields small, informative sets; a poor model yields large, valid but unhelpful sets.

Misconception 3: "Probability calibration and conformal calibration are the same thing"

These are different operations with different goals.
Probability calibration adjusts a model's predicted probabilities so that they better match empirical frequencies (e.g., via Platt scaling or isotonic regression).
Conformal calibration uses a held-out dataset to compute a score threshold that controls coverage for prediction sets. Conformal prediction can use an uncalibrated model and still achieve the coverage guarantee; conversely, a perfectly calibrated model does not automatically produce valid prediction sets without the conformal construction.

Misconception 4: "Validity implies efficiency"

A prediction method can be valid (achieving the desired coverage) but completely uninformative. For example, always returning the entire label set $\mathcal{Y}$ guarantees perfect coverage in classification, but provides no useful information about the true label.
Validity is a *minimum requirement*; efficiency (i.e., how small and informative the prediction sets are) is what makes conformal prediction useful in practice.

7 Final Remarks

7.1 Strengths of Conformal Prediction

Conformal prediction has several properties that distinguish it from other uncertainty quantification approaches:

- **Distribution-free.** The coverage guarantee requires only exchangeability of the data, not any parametric distributional assumption. This makes conformal prediction applicable even when the data-generating distribution is unknown or complex.
- **Model-agnostic.** The procedure wraps around any pre-trained model. Neural networks, gradient-boosted trees, linear models, and even black-box systems can serve as the underlying predictor without modification.
- **Finite-sample validity.** Unlike asymptotic methods that only become valid as $n \to \infty$, conformal prediction provides exact finite-sample guarantees for any calibration set size $n_{\text{cal}} \geq 1$.
- **Simplicity.** The calibration step requires only sorting a list of numbers and computing a quantile, operations that are computationally trivial regardless of dataset size.

7.2 Limitations and Extensions

No method is without limitations. The main caveats of conformal prediction are:

- **Marginal, not conditional, coverage.** The guarantee is averaged over all test inputs. For specific subpopulations or input regions, the empirical coverage may be higher or lower than $1 - \alpha$.
- **Exchangeability Assumption.** This is the backbone of the framework. It assumes the order of data points does not matter. In practice, this is violated if:
  - There is **Data Drift**: the distribution of the test data differs from the calibration data.
  - There is **Temporal Dependency**: in time-series (e.g., financial markets or weather sensors), the recent past is usually more relevant than the distant past.

  If the environment changes (e.g., a new economic regime), the $\hat{q}$ threshold calculated on old data will no longer provide valid coverage.
- **Efficiency depends on the score.** The coverage guarantee always holds, but the prediction sets may be unnecessarily large if the nonconformity score is poorly designed. Improving the score improves efficiency without affecting validity.

Several important extensions of the basic framework exist:

- **Cross-conformal prediction**: replaces the train/calibration split with a cross-validation-like procedure, avoiding the loss of training data.
- **Mondrian conformal prediction**: conditions on class or group labels to provide class-conditional (rather than marginal) coverage guarantees.

- **Online conformal prediction**: extends the framework to sequential data streams where the data may not be exchangeable.
- **Risk control with conformal prediction**: generalizes from coverage to arbitrary bounded loss functions, enabling control over metrics such as false discovery rate or recall.

7.3 Putting It All Together: Validity vs. Efficiency

Suppose two conformal classifiers both target 90% coverage ($\alpha = 0.1$).
Classifier A returns the full set of 10 classes for almost every input. On 1,000 test examples, the true label is included in 1,000 out of 1,000 cases (100% empirical coverage), well above the 90% guarantee.
Classifier B usually returns 1 or 2 classes, and occasionally returns 3 or 4 classes for ambiguous inputs. On 1,000 test examples, the true label is included in 912 out of 1,000 cases (91.2% empirical coverage), also above the 90% guarantee.
Both classifiers are *valid*. However, Classifier B is more *efficient*: its prediction sets are smaller and therefore more informative. Classifier A's sets are safe but useless; they provide no discrimination between labels.
This illustrates the central design goal in conformal prediction:

- **Validity** is the minimum requirement and is guaranteed by the conformal construction.
- **Efficiency** (small prediction sets) is what makes the method useful, and it is earned through the quality of the underlying model and the nonconformity score.

7.4 Practical Guidance

Guidelines for practitioners

- Always use a **held-out calibration set** that was not used during model training. Reusing training data for calibration invalidates the coverage guarantee.
- Choose α to reflect the practical cost of errors in the application. In safety-critical settings (medical, legal, autonomous systems), α should be very small (e.g., $\alpha = 0.01$); in exploratory settings, a looser guarantee (e.g., $\alpha = 0.2$) may be acceptable.
- Invest in a good **nonconformity score**. The coverage guarantee is free; the efficiency (smallness of the prediction sets) must be earned through score design.
- Check empirical coverage on held-out data as a sanity check. If the observed coverage deviates substantially from $1 - \alpha$, investigate whether the exchangeability assumption is violated.

7.5 Key Takeaways

Summary

- Conformal prediction wraps around any model to produce prediction sets (classification) or prediction intervals (regression) with a guaranteed coverage of at least $1 - \alpha$.
- The only requirement is that calibration and test data be exchangeable; no distributional assumptions are needed.
- The procedure has three main stages: (1) train the base model on $\mathcal{D}_{\text{train}}$, (2) compute nonconformity scores on a held-out calibration set $\mathcal{D}_{\text{cal}}$ and extract the empirical quantile $\hat{q}$, (3) at test time, include any class (or interval endpoint) whose score is at most $\hat{q}$.
- The choice of nonconformity score does not affect validity but strongly affects efficiency. Adaptive scores produce smaller, more informative sets.
- For regression, Conformalized Quantile Regression (CQR) combines quantile regression with the conformal step to produce intervals that are both valid and adaptive.
- Validity and efficiency are distinct goals. A valid method that always returns the full label set is useless; a good method is both valid and efficient.

8 Further Reading

The topic of conformal prediction connects ideas from machine learning, statistics, and uncertainty quantification. The references below provide complementary perspectives, ranging from introductory material to foundational theory.

- *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification* [Angelopoulos and Bates, 2021].

  This is an excellent starting point for learning conformal prediction. The paper presents the main ideas in a clear and intuitive way, with practical examples and minimal technical overhead. It covers both classification and regression, and emphasizes the key idea of distribution-free coverage guarantees.

- *Algorithmic Learning in a Random World* [Vovk et al., 2005].

  This book is the foundational reference for conformal prediction. It develops the theory in depth, including the role of exchangeability and the formal guarantees behind the method. While more mathematically involved, it provides the conceptual underpinnings of the framework.

- *Conformalized Quantile Regression* [Romano et al., 2019].

  This paper introduces Conformalized Quantile Regression (CQR), a powerful method for regression problems. It combines quantile regression with conformal calibration to produce adaptive prediction intervals with finite-sample coverage guarantees. It is particularly relevant for modern machine learning applications.

- *Regression Quantiles.* [Koenker and Bassett, 1978]

This classical paper introduces quantile regression, a key building block for methods such as CQR. It provides a way to estimate conditional quantiles of the response variable, enabling more flexible modeling of uncertainty than mean regression.

Note

A recommended reading path is:

1. Start with Angelopoulos and Bates [2021] for intuition
2. Explore Romano et al. [2019] for modern regression methods
3. Consult Vovk et al. [2005] for theoretical foundations

8 References

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021. URL `https://arxiv.org/abs/2107.07511`.

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Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, pages 3543–3553, 2019.

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